Project 2

September 25, 2026 by 5 p.m.

Complete each of the following problems. To submit this project, you must type your solutions using your preferred software (e.g., Word, LaTeX, or Typst) and submit your work as a PDF. For each problem, you must show your work in order to receive credit. No credit will be received if you do not show your work. You may include scans of handwritten work, but the final solution must be typed.

A food science research team is investigating the microbial growth of E. coli. The bacteria cultures are prepared from the same initial stock and plated onto identical growth media. Each culture is placed in its own incubation unit. Although the incubation units are set to the optimal growth temperature, the variation of each machine means that the temperature in °C, is a random variable. A tolerance of α is set so that each machine guarantees that the temperature does not go below $(-\alpha)$°C or above α°C. For our convenience, the distribution of temperature, T, was shifted so that it is symmetric about zero with a support of $t \in \{-\alpha, \alpha\}$. It is reasonable to assume that the temperature of each machine is held constant. The pdf for T is given below.

$$f_T(t) = c \cdot \left(1 - \left(\frac{t}{\alpha}\right)^2\right); \quad -\alpha \leq t \leq \alpha$$

1. Find the value of c that produces a valid pdf for temperature, for any given α.
2. Compute $E[T]$.
3. Compute $V[T]$.
4. Your colleague decides to use Fahrenheit in their report, which is computed as $°F = \frac{9}{5}°C + 32$. Help them with their calculations by computing the expectation and variance of T in °F.

Previous research established that E. coli doubles approximately every 20 minutes when grown under optimal conditions. Knowing that growth will decay when the temperature deviates from the optimal temperature, a parabolic distribution is chosen for $R|T = t$ is is provided below.

$$f_{R|T=t}(r|T = t) = c_R \left[1 - \left(\frac{r - \mu_R(t)}{\gamma}\right)^2\right]; \quad \mu_R(t) - \gamma \leq r \leq \mu_R(t) + \gamma$$

Let $\mu_R(t) = \frac{\ln 2}{20} - \beta(t)^2$ be the expected growth rate at a given temperature, t, β be the rate of decay (constant), γ be the temperature difference from $\mu_R(t)$ at which E. coli stops growing (also constant), and c_R be the normalizing constant such that $\int_{-\infty}^{\infty} f_{R|T=t}(r|T = t) = 1$ (you do not need to calculate c_R).

5. Find the joint distribution function of T and R, $f_{T,R}(t, r)$.
6. Find the marginal distribution function of R, $f_R(R)$
7. Find the cumulative distribution function of R, $F_R(r)$
8. Find the covariance of T and R, $\text{Cov}(T, R)$
9. Find the correlation of T and R, $Corr(T, R)$
10. Are T and R independent? Prove this by showing the (in)equality relationship between $f_{T,R}(t, r)$ and $f_T(t)f_R(r)$.